

Exp No:3.a Conduct an experiment to show data visualization using line plot.

AIM:

Conduct an experiment to show data visualization using line plot.

Description:

Take any sample data either through csv file or data fetched directly through code.

CODE:

```
import matplotlib.pyplot as plt

Overs = list(range(5, 51, 5))

India_Score = [28, 60, 95, 135, 175, 210, 245, 280, 320, 355]

Australia_Score = [22, 55, 88, 120, 160, 195, 225, 255, 290, 315]

plt.plot(Overs, India_Score, color="blue", label="INDIA")

plt.plot(Overs, Australia_Score, color="orange", label="AUSTRALIA")

plt.title("INDIA Vs AUSTRALIA")

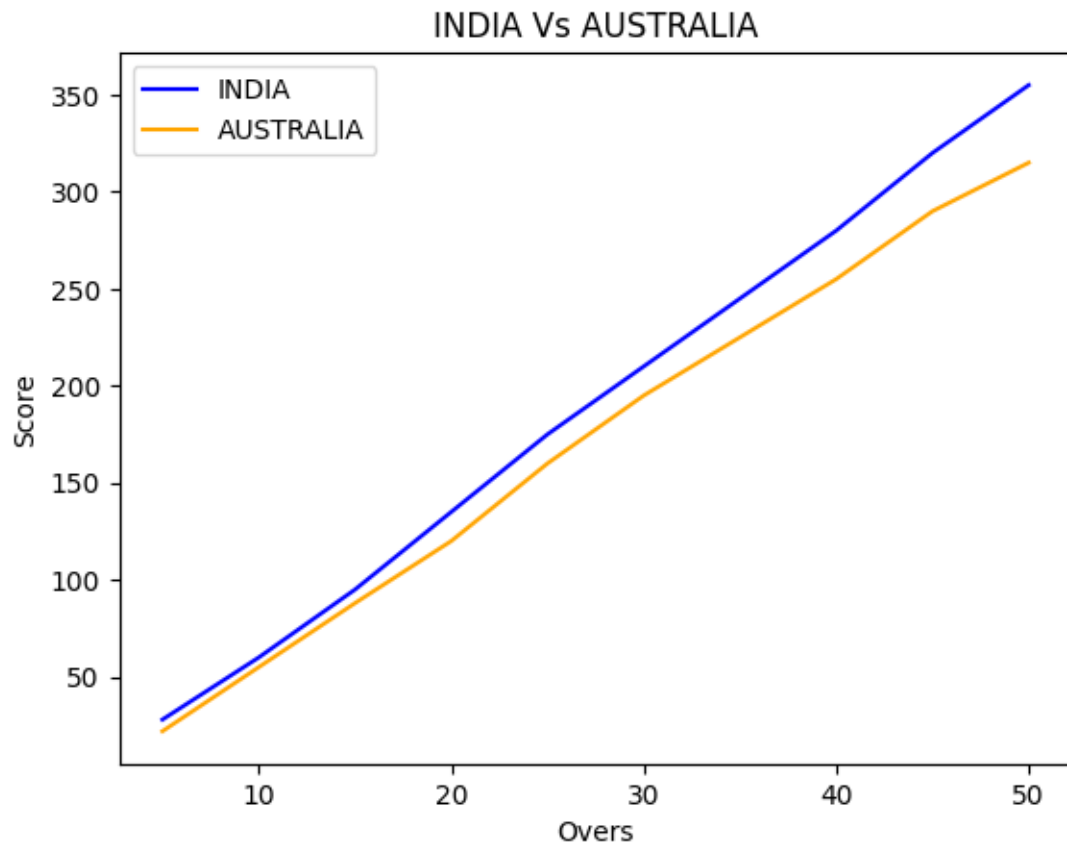
plt.xlabel("Overs")

plt.ylabel("Score")

plt.legend(loc="upper left")

plt.show()
```

OUTPUT:



RESULT:

The line plot was successfully generated showing the score comparison between india and Australia over 50 years. The graph clearly illustrate the scoring trends of both teams.

Exp No:3.b Conduct an experiment to show data visualization using bar chart.

AIM:

Conduct an experiment to show data visualization using bar chart.

Description:

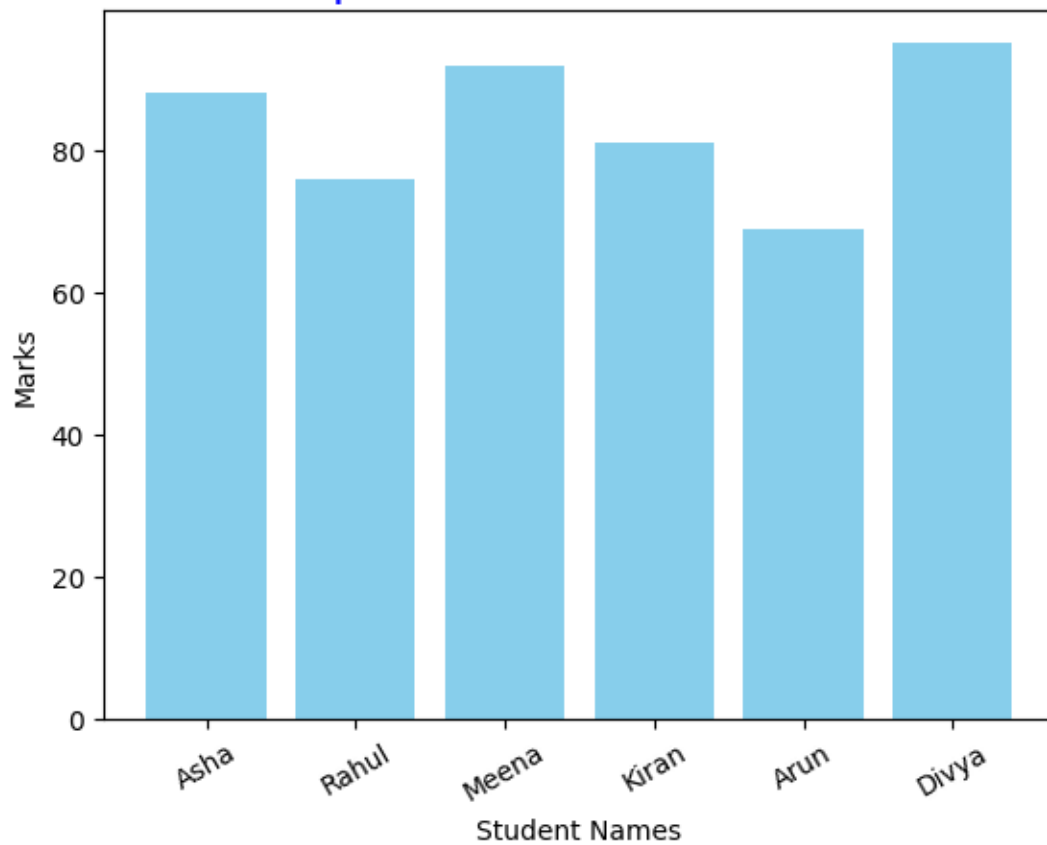
Take any sample data either through csv file or data fetched directly through code.

CODE:

```
import matplotlib.pyplot as plt
import numpy as np
Names = ['Asha', 'Rahul', 'Meena', 'Kiran', 'Arun', 'Divya']
Marks = [88, 76, 92, 81, 69, 95]
xaxis = np.arange(len(Names))
plt.bar(Names, Marks, color='skyblue')
plt.xticks(xaxis, Names, rotation=30)
plt.xlabel("Student Names")
plt.ylabel("Marks")
plt.title("Comparison of Student Marks", fontsize=18, color="blue")
plt.show()
```

OUTPUT:

Comparison of Student Marks



RESULT:

The bar chart was successfully created using the given sample data. It clearly compares the marks obtained by students, making the data easy to analyse and interpret usually.

Exp No:3.c Conduct an experiment to show data visualization using pie chart.

AIM:

Conduct an experiment to show data visualization using pie chart.

Description:

Take any sample data either through csv file or data fetched directly through code.

CODE:

```
import matplotlib.pyplot as plt

labels = ['Maths', 'Science', 'English', 'Tamil']
marks = [90, 75, 65, 70]
colors = ['blue', 'green', 'orange', 'red']
explode = (0.1, 0, 0, 0)

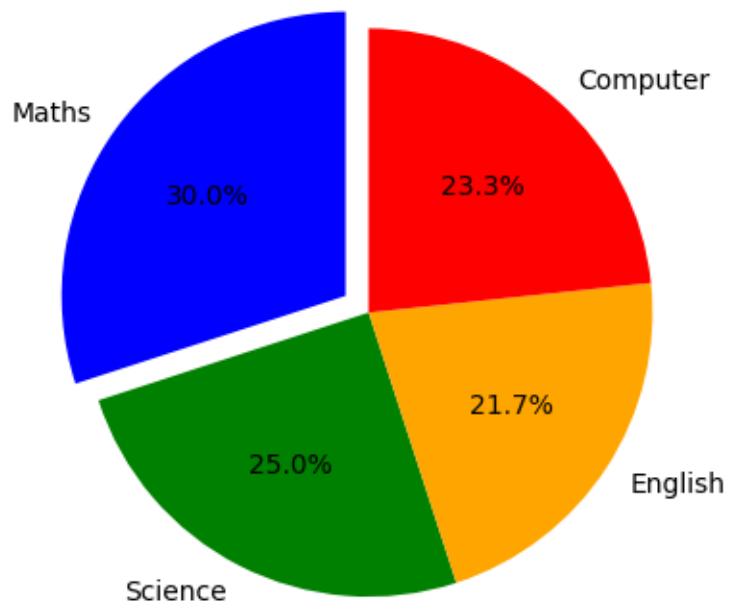
plt.pie(marks,
        labels=labels,
        colors=colors,
        explode=explode,
        autopct='%1.1f%%',
        startangle=90)

plt.title("Student Marks Distribution")

plt.show()
```

OUTPUT:

Student Marks Distribution



RESULT:

The experiment was successfully completed and the pie chart clearly represents the given data using proper labels, colors, proportions.